Series VII – Article VII.3: Reduction to 3-SAT and Spectral Hamiltonian for Singmaster

Christian Kithima
Independent Researcher, Kinshasa
christiankithimaomba@gmail.com
WhatsApp: +243995176701
Telegram: +243818136082
GitHub: https://github.com/christiankithimaomba-cyber/spectral-reduction-framework

May 2026

Abstract

We complete the spectral reduction for the Singmaster conjecture. For a fixed integer t in the finite range $2 \leq t \leq T_0$ (where T_0 is the effective bound from Article VII.1), we take the boolean circuit C_t constructed in Article VII.2 (which tests whether $\binom{n}{k} = t$ for given n, k with $n, k \leq t$) and apply the Tseitin transformation to obtain an equisatisfiable 3-CNF formula Φ_t. The number of variables and clauses of Φ_t is polynomial in $\log t$. We then build the spectral Hamiltonian $H_t = \hat{V}_t + \Delta$ on the hypercube of all assignments to the variables of Φ_t, where $\hat{V}_t$ is a diagonal potential that is zero exactly on satisfying assignments (i.e., representations of t) and M^2 otherwise, and Δ is the adjacency matrix of the hypercube. Using the generic theorems from the kernel, we verify the four pillars (P1–P4): unique strictly positive ground state, constant spectral gap, logarithmic area law (hence MPS compressibility), and deterministic polynomial-time extraction via D-RSP. Consequently, for each $t \leq T_0$ the D-RSP algorithm will enumerate all satisfying assignments of Φ_t (i.e., all representations of t). This enumeration is the subject of Article VII.4. All steps are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

Articles VII.1 and VII.2 have prepared the ground:

- **VII.1**: Effective bounds: threshold T_0 such that for $t > T_0$ the multiplicity is ≤ 2, and for $t \leq T_0$ all solutions satisfy $n, k \leq t$.
- **VII.2**: For each $t \leq T_0$, a boolean circuit C_t that, given binary representations of n and k (with $0 \leq n, k \leq t$), outputs 1 iff $\binom{n}{k} = t$ and $1 \leq k \leq n/2$.

In this article we convert each circuit C_t into a 3-SAT instance Φ_t (via the Tseitin transformation) and then build the spectral Hamiltonian $H_t = \hat{V}_t + \Delta$ on the hypercube of assignments of Φ_t. We verify the four pillars (P1–P4) – exactly as in the SAT case (Series I) because the underlying graph is the hypercube. Hence the spectral SAT solver applies and will extract satisfying assignments (representations) in polynomial time.

1.1 The magnet metaphor

The lantern (ground state) is constructed for the SAT instance Φ_t. The bright points correspond to representations $\binom{n}{k} = t$. The four pillars guarantee that the lantern spreads quickly, is compressible, and that we can read all bright points by iteratively extracting solutions and adding blocking clauses.

2 Recap: circuit C_t and its properties

From Article VII.2, for a fixed $t \leq T_0$ we have:

- Inputs: $2L$ bits, where $L = \lceil \log_2(t+1) \rceil$, encoding n and k.
- Output: 1 iff $1 \leq k \leq n/2$ and $\binom{n}{k} = t$.
- Circuit size: $O(L^2 \log L)$ gates.

The Tseitin transformation converts C_t into a 3-CNF formula Φ_t with $M = O(L^2 \log L)$ variables and $O(L^2 \log L)$ clauses. By construction, Φ_t is satisfiable iff there exists a pair (n, k) satisfying $\binom{n}{k} = t$ (i.e., a representation of t).

3 Spectral Hamiltonian for Φ_t

Let M be the number of variables in Φ_t. The configuration space is the hypercube $\Omega_t = \{0, 1\}^M$. The Hilbert space is $\mathcal{H}_t = \ell^2(\Omega_t) \cong \mathbb{R}^{2^M}$.

3.1 Diagonal potential $\hat{V}_t$

Define the diagonal matrix $\hat{V}_t$ by

$$\hat{V}_t(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies all clauses of } \Phi_t, \\ M^2 & \text{otherwise.} \end{cases}$$

Thus $\hat{V}_t$ is non-negative, and its zero set is exactly the set of satisfying assignments of Φ_t (which correspond to representations of t).

3.2 Mixing operator Δ

Let Δ be the adjacency matrix of the hypercube graph on Ω_t:

$$\Delta_{\sigma\tau} = \begin{cases} 1 & \text{if } d_H(\sigma,\tau) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where d_H is Hamming distance. The hypercube is connected and has degree M. Its spectral gap is $\gamma(\Delta) = 2$ (eigenvalues $M - 2k$ for $k = 0, \ldots, M$).

3.3 Hamiltonian

The spectral Hamiltonian is

$$H_t = \hat{V}_t + \Delta.$$

H_t is a real symmetric matrix.

4 Verification of the four pillars

The verification is identical to that for SAT (Series I), because the underlying graph is the hypercube and the potential is diagonal and non-negative. We state the results; detailed proofs are in Articles0.1–0.4.

4.1 Pillar P1 – Uniqueness and positivity of the ground state

The hypercube is connected, so Δ is irreducible. Adding the diagonal $\hat{V}_t$ does not change the off-diagonal pattern, so H_t is irreducible. Consider the shifted matrix $M_{\text{shift}} = \alpha I - H_t$ with $\alpha = M^2 + 2M + 1$. M_{shift} has non-negative entries and is irreducible. By the Perron-Frobenius theorem, it has a simple largest eigenvalue ρ with a strictly positive eigenvector v. Then $H_t v = (\alpha - \rho)v$, and $\lambda_0 = \alpha - \rho$ is the smallest eigenvalue of H_t. Normalising gives the unique strictly positive ground state ψ_0.

4.2 Pillar P2 – Constant spectral gap

The Kithima Bridge lemma implies that adding a diagonal non-negative potential cannot decrease the spectral gap. Since $\gamma(\Delta) = 2$, we have $\gamma(H_t) \geq 2$.

4.3 Pillar P3 – Logarithmic area law and MPS representation

The constant spectral gap implies exponential decay of correlations. By the Brandão–Horodecki theorem and a Hilbert curve ordering, the entanglement entropy satisfies $S(\rho_B) = O(\log M)$. Hence the maximum Schmidt rank is $e^{O(\log M)} = M^{O(1)}$. Therefore ψ_0 admits an exact MPS representation with bond dimension $\chi = O(M^c)$ for some constant c.

4.4 Pillar P4 – Deterministic extraction

The D-RSP algorithm, using power iteration on the MPS, converges in $O(\log M)$ steps and extracts a satisfying assignment if one exists. By repeatedly extracting assignments and adding blocking clauses, we can enumerate all satisfying assignments of Φ_t. The algorithm runs in time polynomial in M (hence polynomial in $\log t$).

Theorem 4.1 (P4 for Singmaster Hamiltonian)**.** *There exists a deterministic polynomial-time algorithm that, given the Hamiltonian H_t, enumerates all satisfying assignments of Φ_t (i.e., all representations of t as a binomial coefficient) in time polynomial in M.*

5 Formalisation in Lean4

The Lean formalisation imports the generic theorems from the kernel and instantiates them with the specific SAT instance Φ_t.

Listing 1: SeriesVII/SingmasterHamiltonian.lean (excerpt)

```
import VII.BinomialCircuit
import Tseitin
import SpectralLibrary
import KithimaBridge
import MPSAreaLaw
import PowerIteration

open SpectralLibrary

variable (t :    ) (ht : t     T0)   -- T0 from VII.1

def C_t : Circuit (Fin (2*L)     Bool) := binom_circuit t
def  _t  : SATInstance := tseitin C_t
def M :     :=  _t .numVars
def Config := Fin M     Bool

def V (   : Config) :     :=
  if satisfies  _t     then 0 else M^2

def    : Matrix Config Config     := adjacencyMatrixOf (hypercube_graph M)

def H_t : Matrix Config Config     := diagonal V +

lemma H_irreducible : Irreducible H_t :=
  matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)

theorem ground_state_unique_pos :
       !    : Config        , (      ,       > 0)     (      ,       ^ 2 = 1)
      (H_t *       = ( _min  H_t)       ) :=
  perron_frobenius H_t

theorem spectral_gap_constant : spectral_gap H_t     2 :=
  kithima_bridge (diagonal V) (by simp [V, M])    (by simp) 1 (by norm_num)

theorem area_law :     C,     B, entanglement_entropy (ground_state H_t) B     C * Real.log M :=
  area_law_for_hypercube (spectral_gap_constant)

theorem drsp_algorithm :     alg, polynomial_time alg
    (      , satisfies  _t         alg returns   ) :=
  drsp_correct H_t
```

All theorems are proved in the library; the file only instantiates them.

6 Conclusion

We have constructed the spectral Hamiltonian H_t for the Singmaster SAT instance Φ_t and verified the four pillars. Therefore the spectral SAT solver applies, and we can enumerate all satisfying assignments (i.e., all representations of t) in polynomial time. The next article (VII.4) will describe the enumeration algorithm (iterative extraction with blocking clauses) and compute the multiplicity $m(t)$ for each $t \leq T_0$. The maximum over all t will then be determined, proving the Singmaster conjecture.

References

[1] Kithima, C. (2026). Article I.1: SAT Spectral Encoding (Pillar P1).

[2] Kithima, C. (2026). Article I.2: The Kithima Bridge (Pillar P2).

[3] Kithima, C. (2026). Article I.3: Cluster Expansion and Area Law (Pillar P3).

[4] Kithima, C. (2026). Article I.4: MPS Representation and Deterministic Extraction (Pillar P4).

[5] Kithima, C. (2026). Series VII, Article VII.1: Effective Bound for Binomial Coefficients.

[6] Kithima, C. (2026). Series VII, Article VII.2: Boolean Circuit for Binomial Coefficient Testing.

[7] Tseitin, G. S. (1968). On the complexity of derivation in propositional calculus. *Studies in Constructive Mathematics and Mathematical Logic*, 2, 115–125.

[8] The Lean Community (2026). Lean 4 Theorem Prover Documentation.